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EXERCISE 1

Q1. Create all the above tables using appropriate DDL commands by defining suitable data types and integrity constraints such as Primary Key, Foreign Key, NOT NULL and UNIQUE wherever applicable. After successfully creating the schema, insert all the records provided in the tables above. Describe the table and display the entire table with all inserted records.

CODE:

DESCRIBE COURSE ;

DESCRIBE DEPARTMENT ;

DESCRIBE ENROLLEMNT ;

DESCRIBE FACULTY;

DESCRIBE STUDENT;

SELECT * FROM COURSE;

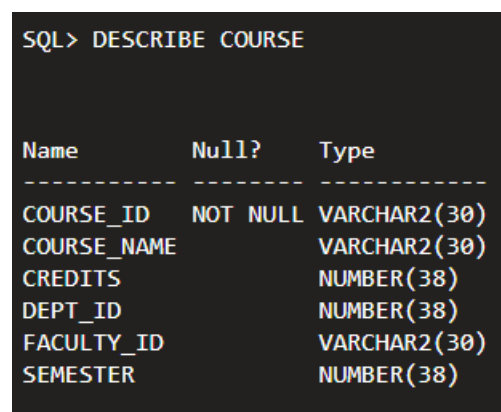
SELECT * FROM DEPARTMENT;

SELECT * FROM ENROLLEMNT;

SELECT * FROM FACULTY;

SELECT * FROM STUDENT;

OUTPUT:

A screenshot of a SQL command prompt showing the output of the 'DESCRIBE COURSE' command. The output is a table with three columns: Name, Null?, and Type. The rows list the columns of the COURSE table: COURSE_ID (NOT NULL, VARCHAR2(30)), COURSE_NAME (VARCHAR2(30)), CREDITS (NUMBER(38)), DEPT_ID (NUMBER(38)), FACULTY_ID (VARCHAR2(30)), and SEMESTER (NUMBER(38)).

Name	Null?	Type
COURSE_ID	NOT NULL	VARCHAR2(30)
COURSE_NAME		VARCHAR2(30)
CREDITS		NUMBER(38)
DEPT_ID		NUMBER(38)
FACULTY_ID		VARCHAR2(30)
SEMESTER		NUMBER(38)

```
SQL> DESCRIBE DEPARTMENT
```

Name	Null?	Type
DEPT_ID	NOT NULL	NUMBER(38)
DEPARTMENT_NAME		VARCHAR2(30)
HOD_NAME		VARCHAR2(30)
OFFICE_LOCATION		VARCHAR2(30)
BUDGET		NUMBER(38)

```
SQL> DESCRIBE ENROLLEMNT
```

Name	Null?	Type
ENROLLMENT_ID	NOT NULL	VARCHAR2(30)
STUDENT_ID		VARCHAR2(30)
COURSE_ID		VARCHAR2(30)
ENROLLEMENT_DATES		VARCHAR2(30)
GRADE		VARCHAR2(30)

```
SQL> DESCRIBE FACULTY
```

Name	Null?	Type
FACULTY_ID	NOT NULL	VARCHAR2(30)
FACULTY_NAME		VARCHAR2(30)
DESIGNATION		VARCHAR2(30)
DEPT_ID		NUMBER(38)
DOJ		VARCHAR2(30)
SALARY		NUMBER(38)

```
SQL> DESCRIBE STUDENT
```

Name	Null?	Type
STUDENT_ID	NOT NULL	VARCHAR2(30)
STUDENT_NAME		VARCHAR2(30)
DOB		VARCHAR2(30)
GENDER		VARCHAR2(30)
DEPT_ID		NUMBER(38)
SEMESTER		NUMBER(38)
CGPA		NUMBER(3,2)

	COURSE_ID	COURSE_NAME	CREDITS	DEPT_ID	FACULTY_ID	SEMESTER
1	C101	Database Systems	4	101	F101	5
2	C102	Operating Systems	4	101	F101	5
3	C103	Computer Networks	4	102	F102	5
4	C104	Digital Electronics	3	103	F103	5
5	C105	Thermodynamics	4	104	F104	5
6	C106	Structural Analysis	4	105	F105	5
7	C107	Power Systems	3	106	F106	5
8	C108	Bioinformatics	3	107	F107	5
9	C109	Machine Learning	4	108	F108	5
10	C110	Data Mining	4	109	F109	5
11	C111	Linear Algebra	3	110	F110	3
12	C112	Artificial Intelligence	4	108	F108	3

	DEPT_ID	DEPARTMENT_NAME	HOD_NAME	OFFICE_LOCATION	BUDGET
1	101	Computer Science	Dr. R. Kumar	AB-1 Block	2500000
2	102	Information Technology	Dr. S. Mehta	AB-2 Block	2200000
3	103	Electronics COMMUN	Dr. P. Nair	AB-3 Block	2600000
4	104	Mechanical Engineer	Dr. A. Sharma	MB-1 Block	2800000
5	105	Civil Engineering	Dr. D. Singh	CB-1 Block	2000000
6	106	Electrical Engineering	Dr. K. Rao	EB-2 Block	2400000
7	107	Biotechnology	Dr. M. Joseph	BT Block	2100000
8	108	Artificial Intelligence	Dr. N. Das	AI Block	3000000
9	109	Data Science	Dr. T. Patel	DS Block	3200000
10	110	Mathematics	Dr. G. Roy	Science Block	1800000

	ENROLLMENT_ID	STUDENT_ID	COURSE_ID	ENROLLEMENT_DATE	GRADE
1	E101	S101	C101	15-Jul-25	A
2	E102	S102	C103	15-Jul-25	A+
3	E103	S103	C104	15-Jul-25	B+
4	E104	S104	C105	15-Jul-25	A
5	E105	S105	C106	15-Jul-25	B
6	E106	S106	C107	15-Jul-25	A+
7	E107	S107	C108	15-Jul-25	A
8	E108	S108	C109	15-Jul-25	O
9	E109	S109	C110	15-Jul-25	A
10	E110	S110	C111	15-Jul-25	B+
11	E111	S111	C102	15-Jul-25	A
12	E112	S112	C112	15-Jul-25	O

	FACULTY_ID	FACULTY_NAME	DESIGNATION	DEPT_ID	DOJ	SALARY
1	F101	Dr. Anita Sharma	Professor	101	15-Jul-12	145000
2	F102	Dr. Vivek Rao	Professor	102	11-Jan-14	142000
3	F103	Dr. Rahul Das	Associate Professor	103	20-Jun-16	118000
4	F104	Dr. Priya Nair	Assistant Professor	104	18-May-19	82000
5	F105	Dr. Arvind Kumar	Professor	105	14-Aug-13	138000
6	F106	Dr. Sneha Gupta	Associate Professor	106	01-Apr-17	112000
7	F107	Dr. Ramesh Patel	Assistant Professor	107	09-Nov-20	78000
8	F108	Dr. Neha Singh	Professor	108	15-Feb-11	155000
9	F109	Dr. Mohan Roy	Associate Professor	109	30-Sep-18	110000
10	F110	Dr. Kavita Bose	Assistant Professor	110	21-Jul-21	76000

	STUDENT_ID	STUDENT_NAME	DOB	GENDER	DEPT_ID	SEMESTER	CGPA
1	S101	Aditya Sharma	15-May-04	M	101	5	8.9
2	S102	Harini Subramanian	22-Aug-04	F	102	5	9.12
3	S103	Arindam Chatterjee	12-Feb-03	M	103	7	8.35
4	S104	Mehak Kaur	30-Oct-04	F	104	5	8.02
5	S105	Rohan Deshmukh	18-Jan-03	M	105	7	7.88
6	S106	Ananya Mishra	09-Sep-04	F	106	5	9.35
7	S107	Tashi Bhutia	16-Jun-03	M	107	7	8.41
8	S108	Nandini Reddy	05-Mar-04	F	108	5	9.48
9	S109	Devansh Patel	28-Apr-04	M	109	5	8.66
10	S110	Isha Boro	01-Dec-03	F	110	7	8.72
11	S111	Rahul Nayak	19-Jul-04	M	101	3	7.95
12	S112	Kavya Menon	11-Nov-04	F	108	3	9.21

Q2. Retrieve the Student ID, Student Name, Department Name, and Semester of all students who belong to either Computer Science or Information Technology departments and whose CGPA is greater than or equal to 8.20. Display the output in descending order of CGPA.

CODE:

```
SELECT STUDENT.STUDENT_ID,
       STUDENT.STUDENT_NAME,
       DEPARTMENT.DEPARTMENT_NAME,
       STUDENT.SEMESTER
FROM STUDENT
INNER JOIN DEPARTMENT
ON STUDENT.DEPT_ID = DEPARTMENT.DEPT_ID
```

```

WHERE (DEPARTMENT.DEPARTMENT_NAME = 'Computer Science'

      OR DEPARTMENT.DEPARTMENT_NAME = 'Information Technology' AND
STUDENT.CGPA >= 8.20)

ORDER BY STUDENT.CGPA DESC;

```

OUTPUT:

	STUDENT_ID	STUDENT_NAME	DEPARTMENT_NAME	SEMESTER
1	S102	Harini Subramanian	Information Technology	5
2	S101	Aditya Sharma	Computer Science	5
3	S111	Rahul Nayak	Computer Science	3

QUESTION 3: Display the names of faculty members who are handling more than one course along with the total number of courses assigned to each faculty member.

CODE:

```

SELECT FACULTY.FACULTY_NAME, COUNT(COURSE.COURSE_ID) AS TOTAL_COURSES
FROM FACULTY
INNER JOIN COURSE
ON COURSE.FACULTY_ID = FACULTY.FACULTY_ID
GROUP BY FACULTY.FACULTY_NAME
HAVING COUNT(COURSE.COURSE_ID) > 1;

```

OUTPUT

	FACULTY_NAME	TOTAL_COURSES
1	Dr. Anita Sharma	2
2	Dr. Neha Singh	2

QUESTION 4: Update the semester of all students whose CGPA is greater than 9.00 from Semester 5 to Semester 6. Display the updated records.

CODE:

```
UPDATE STUDENT
```

```
SET STUDENT.SEMESTER = 6
```

```
WHERE (STUDENT.CGPA > 9 AND STUDENT.SEMESTER = 5);
```

```
SELECT * FROM STUDENT;
```

OUTPUT:

	STUDENT_ID	STUDENT_NAME	DOB	GENDER	DEPT_ID	SEMESTER	CGPA
1	S101	Aditya Sharma	15-May-04	M	101	5	8.9
2	S102	Harini Subramanian	22-Aug-04	F	102	6	9.12
3	S103	Arindam Chatterjee	12-Feb-03	M	103	7	8.35
4	S104	Mehak Kaur	30-Oct-04	F	104	5	8.02
5	S105	Rohan Deshmukh	18-Jan-03	M	105	7	7.88
6	S106	Ananya Mishra	09-Sep-04	F	106	6	9.35
7	S107	Tashi Bhutia	16-Jun-03	M	107	7	8.41
8	S108	Nandini Reddy	05-Mar-04	F	108	6	9.48
9	S109	Devansh Patel	28-Apr-04	M	109	5	8.66
10	S110	Isha Boro	01-Dec-03	F	110	7	8.72
11	S111	Rahul Nayak	19-Jul-04	M	101	3	7.95
12	S112	Kavya Menon	11-Nov-04	F	108	3	9.21

QUESTION 5: Add a new column named StudentEmail to the STUDENT table. Populate the email addresses for all students and later modify the column by applying the UNIQUE constraint.

CODE:

```
ALTER TABLE STUDENT
```

```
ADD STUDENT_EMAIL VARCHAR2(40);
```

```
ALTER TABLE STUDENT
```

```
MODIFY STUDENT_EMAIL UNIQUE
```

```
UPDATE STUDENT
```

```
SET STUDENT_EMAIL = STUDENT_NAME || '@vit.ac.in';
```

```
SELECT * FROM STUDENT
```

OUTPUT

	STUDENT_ID	STUDENT_NAME	DOB	GENDER	DEPT_ID	SEMESTER	CGPA	STUDENT_EMAIL
1	S101	Aditya Sharma	15-May-04	M	101	5	8.9	Aditya Sharma@vit.ac.in
2	S102	Harini Subramanian	22-Aug-04	F	102	6	9.12	Harini Subramanian@vit.ac.in
3	S103	Arindam Chatterjee	12-Feb-03	M	103	7	8.35	Arindam Chatterjee@vit.ac.in
4	S104	Mehak Kaur	30-Oct-04	F	104	5	8.02	Mehak Kaur@vit.ac.in
5	S105	Rohan Deshmukh	18-Jan-03	M	105	7	7.88	Rohan Deshmukh@vit.ac.in
6	S106	Ananya Mishra	09-Sep-04	F	106	6	9.35	Ananya Mishra@vit.ac.in
7	S107	Tashi Bhutia	16-Jun-03	M	107	7	8.41	Tashi Bhutia@vit.ac.in
8	S108	Nandini Reddy	05-Mar-04	F	108	6	9.48	Nandini Reddy@vit.ac.in
9	S109	Devansh Patel	28-Apr-04	M	109	5	8.66	Devansh Patel@vit.ac.in
10	S110	Isha Boro	01-Dec-03	F	110	7	8.72	Isha Boro@vit.ac.in
11	S111	Rahul Nayak	19-Jul-04	M	101	3	7.95	Rahul Nayak@vit.ac.in
12	S112	Kavya Menon	11-Nov-04	F	108	3	9.21	Kavya Menon@vit.ac.in

QUESTION 6: Rename the StudentEmail column as UniversityEmail. Verify the changes using DESCRIBE command.

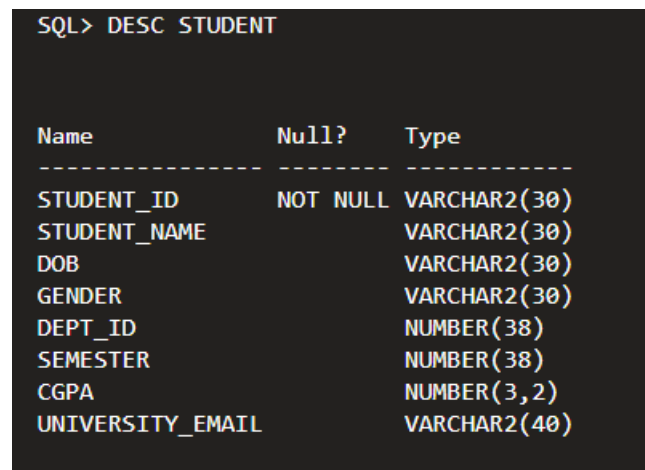
CODE:

```
ALTER TABLE STUDENT
```

```
RENAME COLUMN STUDENT_EMAIL TO UNIVERSITY_EMAIL
```

```
DESC STUDENT
```

OUTPUT:



```
SQL> DESC STUDENT
```

Name	Null?	Type
STUDENT_ID	NOT NULL	VARCHAR2(30)
STUDENT_NAME		VARCHAR2(30)
DOB		VARCHAR2(30)
GENDER		VARCHAR2(30)
DEPT_ID		NUMBER(38)
SEMESTER		NUMBER(38)
CGPA		NUMBER(3,2)
UNIVERSITY_EMAIL		VARCHAR2(40)

QUESTION 7. Increase the size of FacultyDesignation from VARCHAR2(20) to VARCHAR2(40) using ALTER TABLE.

CODE:

```
ALTER TABLE FACULTY
```

```
MODIFY DESIGNATION VARCHAR(40)
```

```
DESC FACULTY
```

OUTPUT:


```
SQL> DESC FACULTY
```

Name	Null?	Type
FACULTY_ID	NOT NULL	VARCHAR2(30)
FACULTY_NAME		VARCHAR2(30)
DESIGNATION		VARCHAR2(40)
DEPT_ID		NUMBER(38)
DOJ		VARCHAR2(30)
SALARY		NUMBER(38)

QUESTION 8: Delete the enrollment record of a student whose enrollment date is earlier than '01-JAN-2025'. Display the updated ENROLLMENT table.

CODE : UPDATE ENROLLEMNT

SET ENROLLEMENT_DATES = TO_DATE(ENROLLEMENT_DATES,'DD-Mon-YY');

SELECT * FROM ENROLLEMNT

DELETE FROM ENROLLEMNT WHERE (ENROLLEMENT_DATES < '01-JAN-2025');

OUTPUT

	ENROLLMENT_ID	STUDENT_ID	COURSE_ID	ENROLLEMENT_D/	GRADE
1	E101	S101	C101	15-JUL-25	A
2	E102	S102	C103	15-JUL-25	A+
3	E103	S103	C104	15-JUL-25	B+
4	E104	S104	C105	15-JUL-25	A
5	E105	S105	C106	15-JUL-25	B
6	E106	S106	C107	15-JUL-25	A+
7	E107	S107	C108	15-JUL-25	A
8	E108	S108	C109	15-JUL-25	O
9	E109	S109	C110	15-JUL-25	A
10	E110	S110	C111	15-JUL-25	B+
11	E111	S111	C102	15-JUL-25	A
12	E112	S112	C112	15-JUL-25	O

QUESTION 9: Rename the COURSE table as SUBJECT and verify the schema.

CODE:

```
ALTER TABLE COURSE RENAME TO SUBJECT;
```

```
DESC SUBJECT;
```

OUTPUT:

```
SQL> DESC SUBJECT
```

Name	Null?	Type
COURSE_ID	NOT NULL	VARCHAR2(30)
COURSE_NAME		VARCHAR2(30)
CREDITS		NUMBER(38)
DEPT_ID		NUMBER(38)
FACULTY_ID		VARCHAR2(30)
SEMESTER		NUMBER(38)

QUESTION 10: Remove the UniversityEmail column from STUDENT table and display the modified table structure.

CODE:

```
TABLE STUDENT
```

```
DROP COLUMN UNIVERSITY_EMAIL
```

OUTPUT :

```
SQL> DESC STUDENT
```

Name	Null?	Type
STUDENT_ID	NOT NULL	VARCHAR2(30)
STUDENT_NAME		VARCHAR2(30)
DOB		VARCHAR2(30)
GENDER		VARCHAR2(30)
DEPT_ID		NUMBER(38)
SEMESTER		NUMBER(38)
CGPA		NUMBER(3,2)

EXERCISE 2

QUESTION 1. Modify the schema created in Exercise-1 by incorporating the new relations ATTENDANCE, COURSE_PREREQUISITE and EXAM_RESULT. Define all necessary integrity constraints including Composite Primary Keys, Foreign Keys, CHECK constraints, DEFAULT values and UNIQUE constraints.

Code:

```
CREATE TABLE ATTENDANCE(  
    STUDENT_ID VARCHAR(30) REFERENCES STUDENT(STUDENT_ID),  
    COURSE_ID VARCHAR(30) REFERENCES COURSE(COURSE_ID) ,  
    TOTAL_CLASSES INTEGER,  
    CLASSES_ATTENDED INTEGER,  
    ATTENDANCE_PERCENTAGE NUMBER(4,2) CHECK(ATTENDANCE_PERCENTAGE  
    BETWEEN 0 AND 100),  
    PRIMARY KEY(STUDENT_ID,COURSE_ID)  
);
```

```
INSERT INTO ATTENDANCE VALUES('S101', 'C101', 40, 36, 90);  
INSERT INTO ATTENDANCE VALUES('S102', 'C103', 42, 40, 95.24);  
INSERT INTO ATTENDANCE VALUES('S103', 'C104', 38, 32, 84.21);  
INSERT INTO ATTENDANCE VALUES('S104', 'C105', 40, 34, 85);  
INSERT INTO ATTENDANCE VALUES('S105', 'C106', 42, 30, 71.43);  
INSERT INTO ATTENDANCE VALUES('S106', 'C107', 41, 39, 95.12);  
INSERT INTO ATTENDANCE VALUES('S107', 'C108', 38, 33, 86.84);  
INSERT INTO ATTENDANCE VALUES('S108', 'C109', 40, 39, 97.5);  
INSERT INTO ATTENDANCE VALUES('S109', 'C110', 39, 35, 89.74);  
INSERT INTO ATTENDANCE VALUES('S110', 'C111', 40, 31, 77.5);
```

```
CREATE TABLE EXAM_RESUT(
```

```

STUDENT_ID VARCHAR2(30),
COURSE_ID VARCHAR2(30),
INTERNAL_MARKS INTEGER CHECK (INTERNAL_MARKS BETWEEN 0 AND 50),
EXTERNAL_MARKS INTEGER CHECK (EXTERNAL_MARKS BETWEEN 0 AND 50),
TOTAL_MARKS INTEGER,
RESULT VARCHAR2(30) DEFAULT 'PASS',
PRIMARY KEY(STUDENT_ID,COURSE_ID)
);

```

```

INSERT INTO EXAM_RESULT VALUES('S101', 'C101', 45, 44, 89, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S102', 'C103', 48, 46, 94, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S103', 'C104', 41, 39, 80, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S104', 'C105', 40, 42, 82, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S105', 'C106', 33, 37, 70, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S106', 'C107', 49, 47, 96, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S107', 'C108', 43, 40, 83, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S108', 'C109', 50, 48, 98, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S109', 'C110', 44, 43, 87, 'PASS');
INSERT INTO EXAM_RESULT VALUES('S110', 'C111', 38, 41, 79, 'PASS');

```

```

CREATE TABLE COURSE_PREREQUISITE(
    COURSE_ID VARCHAR2(30) REFERENCES COURSE,
    PREREQUISITE_COURSE VARCHAR2(30) REFERENCES COURSE(COURSE_ID),
    MINIMUM_CGPA NUMBER(2,1) CHECK( MINIMUM_CGPA BETWEEN 0 AND 10),
    PRIMARY KEY (COURSE_ID)
);

```

```

INSERT INTO COURSE_PREREQUISITE VALUES('C102', 'C101', 7.5);

```

```

INSERT INTO COURSE_PREREQUISITE VALUES('C103', 'C101', 7.8);
INSERT INTO COURSE_PREREQUISITE VALUES('C109', 'C101', 8.2);
INSERT INTO COURSE_PREREQUISITE VALUES('C110', 'C109', 8.4);
INSERT INTO COURSE_PREREQUISITE VALUES('C112', 'C109', 8.5);
INSERT INTO COURSE_PREREQUISITE VALUES('C107', 'C104', 7);
INSERT INTO COURSE_PREREQUISITE VALUES('C108', 'C104', 7.2);
INSERT INTO COURSE_PREREQUISITE VALUES('C106', 'C105', 7.5);
INSERT INTO COURSE_PREREQUISITE VALUES('C111', 'C101', 6.5);
INSERT INTO COURSE_PREREQUISITE VALUES('C105', 'C111', 6.8);

```

OUTPUT:

```
SQL> DESC ATTENDANCE
```

Name	Null?	Type
STUDENT_ID	NOT NULL	VARCHAR2(30)
COURSE_ID	NOT NULL	VARCHAR2(30)
TOTAL_CLASSES		NUMBER(38)
CLASSES_ATTENDED		NUMBER(38)
ATTENDANCE_PERCENTAGE		NUMBER(4,2)

```
SQL> DESC COURSE_PREREQUISITE
```

Name	Null?	Type
-----	-----	-----
COURSE_ID	NOT NULL	VARCHAR2(30)
PREREQUISITE_COURESE		VARCHAR2(30)
MINIMUM_CGPA		NUMBER(2,1)

```
SQL> DESC EXAM_RESULT
```

Name	Null?	Type
-----	-----	-----
STUDENT_ID	NOT NULL	VARCHAR2(30)
COURSE_ID	NOT NULL	VARCHAR2(30)
INTERNAL_MARKS		NUMBER(38)
EXTERNAL_MARKS		NUMBER(38)
TOTAL_MARKS		NUMBER(38)
RESULT		VARCHAR2(30)

QUESTION 2: Attempt to insert a student record with a Department ID that does not exist. Observe and explain the integrity constraint violated.

CODE:

```
INSERT INTO STUDENT VALUES('S113','XYZ','20-Feb-06','M','000',4,9,NULL)
```

OUTPUT:

```
SQL> INSERT INTO STUDENT VALUES('S113','XYZ','20-Feb-06','M','000',4,9,NULL)

ORA-02291: integrity constraint (RATUL_PAL2024_SCHEMA_WW6ZB.SYS_C005396272) violated - parent key not found

https://docs.oracle.com/error-help/db/ora-02291/
Error at Line: 4 Column: 0
```

Explanation:

The provided department id should be existing inside the department table, else if no such id exist integrity will be violated and will throw back error

QUESTION 3: Attempt to insert duplicate Course IDs into the COURSE table. Explain the error generated.

CODE:

```
INSERT INTO COURSE VALUES('C101','XYZ',4,101,'F101',6)
```

OUTPUT:

```
SQL> INSERT INTO COURSE VALUES('C101','XYZ',4,101,'F101',6)

ORA-
00001: unique constraint (RATUL_PAL2024_SCHEMA_WW6ZB.SYS_C005430540) violated on table RATUL_PAL2024_SCHEMA_WW6ZB.COURSE columns (COURSE_ID)
ORA-03301: (ORA-00001 details) row with column values (COURSE_ID:'C101') already exists

https://docs.oracle.com/error-help/db/ora-00001/
Error at line: 4 Column: 0
```

EXPLANATION:

Hereby the course id is defined as a primary key which means the data entered into this column must be not null and should strictly be unique. Hence when we try to enter such duplicate value we face error.

QUESTION 4: Delete a department that is referenced by multiple students and courses. Explain the behaviour of the database based on the defined foreign key constraints.

CODE:

```
DELETE FROM DEPARTMENT WHERE DEPT_ID = 101
```

OUTPUT:

```
SQL> DELETE FROM DEPARTMENT WHERE DEPT_ID = 101

ORA-02292: integrity constraint (RATUL_PAL2024_SCHEMA_WW6ZB.SYS_C005430541) violated - child record found
https://docs.oracle.com/error-help/db/ora-02292/
Error at Line: 5 Column: 0
```

EXPLANATION:

When we are trying to delete a data from a column which is a primary key as is linked with other foreign key of different table, deleting the data from such a column rises the Foreign key integrity error.

QUESTION 5: Modify the foreign key on COURSE table using ALTER TABLE by first dropping the existing constraint and recreating it with ON DELETE SET NULL option.

CODE:

```
SELECT column_name,constraint_name FROM user_cons_columns WHERE
table_name='COURSE';
```

```
ALTER TABLE COURSE DROP CONSTRAINTS SYS_C005430541
```

```
ALTER TABLE COURSE DROP CONSTRAINTS SYS_C005430542
```

```
ALTER TABLE COURSE
```

```
ADD CONSTRAINT FK_DEPT_ID
```

```
FOREIGN KEY(DEPT_ID)
```

```
REFERENCES DEPARTMENT
```

```
ON DELETE SET NULL;
```

```
ALTER TABLE COURSE
```



```
ADD CONSTRAINT FK_FACULTY_ID
FOREIGN KEY(FACULTY_ID)
REFERENCES FACULTY
ON DELETE SET NULL;
```

OUTPUT:

	COLUMN_NAME		CONSTRAINT_NAME
1	DEPT_ID	👁	FK_DEPT_ID
2	FACULTY_ID		FK_FACULTY_ID
3	CREDITS		SYS_C005430538
4	SEMESTER		SYS_C005430539
5	COURSE_ID		SYS_C005430540

QUESTION 6: Add CHECK constraint to ensure Attendance Percentage remains between 0 and 100.

```
CODE: ALTER TABLE ATTENDANCE MODIFY ATTENDANCE_PERCENTAGE NUMBER(4,2)
CHECK(ATTENDANCE_PERCENTAGE BETWEEN 0 AND 100);
```

OUTPUT:

	STUDENT_ID	COURSE_ID	TOTAL_CLASSES	CLASSES_ATTEND	ATTENDANCE_PER
1	S101	C101	40	36	90
2	S102	C103	42	40	95.24
3	S103	C104	38	32	84.21
4	S104	C105	40	34	85
5	S105	C106	42	30	71.43
6	S106	C107	41	39	95.12
7	S107	C108	38	33	86.84
8	S108	C109	40	39	97.5
9	S109	C110	39	35	89.74
10	S110	C111	40	31	77.5

QUESTION 7: Modify the Marks column from NUMBER(3) to NUMBER(5,2) without affecting existing records.

CODE:

```
UPDATE EXAM_RESULT SET INTERNAL_MARKS = NULL
```

```
ALTER TABLE EXAM_RESULT MODIFY INTERNAL_MARKS NUMBER(5,2)
```

OUTPUT:

Name	Null?	Type
STUDENT_ID	NOT NULL	VARCHAR2(30)
COURSE_ID	NOT NULL	VARCHAR2(30)
INTERNAL_MARKS		NUMBER(5,2)
EXTERNAL_MARKS		NUMBER(38)
TOTAL_MARKS		NUMBER(38)
RESULT		VARCHAR2(30)

QUESTION 8: Alter the STUDENT table by adding a new column named StudentStatus of datatype VARCHAR2(15). Set its default value to 'ACTIVE'. Verify the behaviour by inserting a new student record without specifying the StudentStatus column and display the inserted record.

CODE:

```
ALTER TABLE STUDENT ADD StudentStatus VARCHAR2(15) DEFAULT 'ACTIVE'
```

INSERT INTO

```
STUDENT(STUDENT_ID,STUDENT_NAME,DOB,GENDER,DEPT_ID,SEMESTER,CGPA)
```

```
VALUES('S113','XYZ','03-Feb-06','M',102,4,9.0);
```

OUTPUT:

	STUDENT_ID	STUDENT_NAME	DOB	GENDER	DEPT_ID	SEMESTER	CGPA	STUDENT_EMAIL	STUDENTSTATUS
1	S113	XYZ	03-Feb-06	M	102	4	9	(null)	ACTIVE
2	S101	Aditya Sharma	15-May-04	M	101	5	8.9	(null)	ACTIVE
3	S102	Harini Subramanian	22-Aug-04	F	102	6	9.12	(null)	ACTIVE
4	S103	Arindam Chatterjee	12-Feb-03	M	103	7	8.35	(null)	ACTIVE
5	S104	Mehak Kaur	30-Oct-04	F	104	5	8.02	(null)	ACTIVE
6	S105	Rohan Deshmukh	18-Jan-03	M	105	7	7.88	(null)	ACTIVE
7	S106	Ananya Mishra	09-Sep-04	F	106	6	9.35	(null)	ACTIVE
8	S107	Tashi Bhutia	16-Jun-03	M	107	7	8.41	(null)	ACTIVE
9	S108	Nandini Reddy	05-Mar-04	F	108	6	9.48	(null)	ACTIVE
10	S109	Devansh Patel	28-Apr-04	M	109	5	8.66	(null)	ACTIVE
11	S110	Isha Boro	01-Dec-03	F	110	7	8.72	(null)	ACTIVE
12	S111	Rahul Nayak	19-Jul-04	M	101	3	7.95	(null)	ACTIVE
13	S112	Kavya Menon	11-Nov-04	F	108	3	9.21	(null)	ACTIVE

QUESTION 9: Rename the ATTENDANCE table as STUDENT_ATTENDANCE and verify the changes.

CODE:

```
ALTER TABLE ATTENDANCE RENAME TO STUDENT_ATTENDANCE
```

OUTPUT:

```
SQL> DESC STUDENT_ATTENDANCE
```

Name	Null?	Type
STUDENT_ID	NOT NULL	VARCHAR2(30)
COURSE_ID	NOT NULL	VARCHAR2(30)
TOTAL_CLASSES		NUMBER(38)
CLASSES_ATTENDED		NUMBER(38)
ATTENDANCE_PERCENTAGE		NUMBER(4,2)

QUESTION 10: Drop the CHECK constraint created in Question 6 and explain its effect by inserting invalid attendance data.

CODE:

```
SELECT column_name,constraint_name
```

```
FROM user_cons_columns
```

```
WHERE table_name='STUDENT_ATTENDANCE';
```

```
ALTER TABLE STUDENT_ATTENDANCE DROP CONSTRAINT SYS_C005434241
```

```
ALTER TABLE STUDENT_ATTENDANCE DROP CONSTRAINT SYS_C005430288
```

```
ALTER TABLE STUDENT_ATTENDANCE DROP CONSTRAINT SYS_C005434384
```

```
ALTER TABLE STUDENT_ATTENDANCE MODIFY ATTENDANCE_PERCENTAGE  
NUMBER(6,2)
```

```
INSERT INTO STUDENT_ATTENDANCE VALUES('S111','C111',50,40,101)
```

```
SELECT * FROM STUDENT_ATTENDANCE
```

OUTPUT:

	STUDENT_ID	COURSE_ID	TOTAL_CLASSES	CLASSES_ATTEND	ATTENDANCE_PER
1	S111	C111	50	40	101
2	S101	C101	40	36	90
3	S102	C103	42	40	95.24
4	S103	C104	38	32	84.21
5	S104	C105	40	34	85
6	S105	C106	42	30	71.43
7	S106	C107	41	39	95.12
8	S107	C108	38	33	86.84
9	S108	C109	40	39	97.5
10	S109	C110	39	35	89.74
11	S110	C111	40	31	77.5

